Self-Supervised Vision Foundation Models: A Study of the DINO Family and Its Adaptation to Visual Place Recognition

Project Report — Part I: Theoretical Foundations
(Chapters 1–3)

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Abstract

This report presents a study of the DINO family of self-supervised vision foundation models (DINO, DINOv2, DINOv3) and their application to the Visual Place Recognition (VPR) task. Part I of the report, contained in this document, provides the theoretical groundwork. Chapter 1 situates DINO within the broader landscape of foundation models and motivates self-supervised learning for computer vision. Chapter 2 presents a detailed, graduate-level treatment of the DINO family, including its architecture, self-distillation objective, centering and sharpening mechanisms for collapse avoidance, multi-crop augmentation, the patch-level iBOT objective, the KoLeo regularizer, and the Gram anchoring technique introduced in DINOv3. Chapter 3 describes the feature-analysis study we conduct on the pretrained DINOv2 backbone, covering qualitative and quantitative probes of its representations; results from this chapter are obtained in Part II of the project and are included here as placeholders. Parts II and III will cover a minimum-viable reimplementation of the DINO training loop, the VPR adaptation experiments, and the empirical comparison of adaptation strategies.

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1 Introduction

1.1 From Supervised to Self-Supervised Vision

For the better part of the past decade, progress in computer vision has been driven by the paradigm of *supervised pretraining*: train a deep network on a large, human-labeled dataset such as ImageNet-1k [Russakovsky et al., 2015] and transfer its weights to downstream tasks. This recipe has been remarkably effective, but it carries three well-known limitations.

1. **Annotation cost.** Obtaining clean, class-balanced labels for millions of images is expensive, slow, and often impossible in specialized domains (medical imaging, remote sensing, industrial inspection).

2. **Label-induced bias.** The learned representation is shaped by the particular taxonomy of the labeling scheme. Features optimized for 1000-way ImageNet classification are not necessarily ideal for, say, segmentation, depth estimation, or instance-level retrieval.

3. **Scale ceiling.** Labeled datasets saturate in size at a scale that is orders of magnitude below the quantity of *unlabeled* visual data available on the Internet.

A parallel development in natural language processing — beginning with word embeddings [Mikolov et al., 2013] and culminating in modern large language models — demonstrated that task-agnostic representations trained with *self-supervised* objectives on massive unlabeled corpora can serve as powerful substrates for nearly any downstream task. The *foundation model* [Bommasani et al., 2021] paradigm formalises this observation: one pretrains a large network on a broad data distribution once, then adapts it to a multitude of downstream tasks with minimal additional compute or labeled data.

Replicating this success in computer vision required two things: a training objective that does not rely on labels, and an architecture that benefits from data at very large scale. Vision Transformers (ViTs) [Dosovitskiy et al., 2021] provided the latter. The former has been the subject of active research, yielding contrastive methods (SimCLR [Chen et al., 2020], MoCo [He et al., 2020]), clustering-based methods (SwAV [Caron et al., 2020]), predictive methods (BYOL [Grill et al., 2020]), and masked-reconstruction methods (MAE [He et al., 2022], BEiT [Bao et al., 2022]). The DINO family [Caron et al., 2021, Oquab et al., 2024, Siméoni et al., 2025] combines elements from several of these and represents, at the time of writing, one of the most successful attempts at a general-purpose, purely self-supervised vision foundation model.

1.2 Why DINO?

Three properties single out the DINO family for our study.

Emergent semantic structure without labels. The original DINO paper [Caron et al., 2021] observed that self-supervised Vision Transformers, trained only with a self-distillation objective, spontaneously develop attention maps that segment the salient object from the background — an *emergent* property not explicitly optimized for. This is not merely cosmetic: it suggests that the learned representation encodes object-level semantics in a form directly accessible to downstream heads without further supervision.

State-of-the-art dense features. DINOv2 [Oquab et al., 2024] and, more markedly, DINOv3 [Siméoni et al., 2025] produce *patch-level* features of exceptional quality. On benchmarks like ADE20K semantic segmentation and NYU depth estimation, a frozen DINOv2 or DINOv3 backbone combined with a simple linear probe already reaches competitive performance — matching or exceeding specialized, fully supervised architectures that were explicitly trained for the task.

A clear evolutionary arc. DINO → DINOv2 → DINOv3 constitutes a cleanly documented research progression: DINO introduced the self-distillation core; DINOv2 added the patch-level iBOT [Zhou et al., 2022] objective, the KoLeo regularizer [Sablayrolles et al., 2019], Sinkhorn–Knopp centering, and scaled the data from ImageNet to 142M curated images (LVD-142M); DINOv3 scaled further to 1.7B images and a 7B-parameter ViT, introduced Gram anchoring to prevent the degradation of patch features during long training, replaced learnable positional embeddings with RoPE, and added a high-resolution refinement phase. Each step addresses a specific, identified weakness of its predecessor, making the family an excellent case study in how foundation models are actually engineered.

1.3 Visual Place Recognition: The Downstream Task

Visual Place Recognition (VPR) is the task of identifying the geographic location of a query image by retrieving the most similar image from a geo-tagged reference database [Lowry et al., 2016]. It is a core component of robot localization, autonomous driving, and augmented reality, and is typically evaluated on large-scale benchmarks such as Pitts30k [Torii et al., 2013], Mapillary Street-Level Sequences (MSLS) [Warburg et al., 2020], Tokyo 24/7 [Torii et al., 2015], and Nordland [Sünderhauf et al., 2013]. The evaluation metric is usually Recall@N: a query is considered correctly localized if at least one of the top-N retrieved references lies within a threshold distance (typically 25 m) of the query's ground-truth pose.

VPR is an attractive downstream task for our study for three reasons:

1. **It stresses semantic robustness.** A correct match must be made despite seasonal changes, illumination shifts, viewpoint variation, and dynamic occluders (pedestrians, vehicles). This is exactly the kind of representation-quality test on which strong pretraining shines.

2. **It has a clean global–local structure.** State-of-the-art VPR systems typically combine a *global descriptor* for coarse retrieval with *local descriptors* for geometric re-ranking. DINO's ability to produce both [CLS]-level global features and high-quality patch-level local features makes it a natural backbone for this structure.

3. **The recent literature is explicitly built around DINO.** Several state-of-the-art VPR methods — SelaVPR [Lu et al., 2024b], SelaVPR++ [Lu et al., 2025], CricaVPR [Lu et al., 2024a], EffoVPR [Tzachor et al., 2024], SALAD [Izquierdo and Civera, 2024] — use DINOv2 as their backbone and differ primarily in how they *adapt* it. This gives us a well-defined comparison space for our experiments.

1.4 Contributions and Report Structure

This project contributes, across its three parts:

(1) A detailed graduate-level exposition of the DINO family's mathematical foundations and training mechanisms (Chapter 2).

(2) An empirical feature-analysis study of the publicly released DINOv2 backbone, covering attention-map visualization, principal-component analysis of patch features, and non-parametric probes such as kNN classification and zero-shot segmentation (Chapter 3).

(3) A minimum-viable from-scratch reimplementation of the DINO self-distillation training loop on a small dataset, serving as a pedagogical demonstration that the core mechanism is understood (Part II, Chapter 4).

(4) A survey and taxonomy of adaptation techniques for vision foundation models (Part II, Chapter 5).

(5) An implementation and empirical comparison of multiple VPR adaptation strategies — a frozen-backbone baseline with GeM pooling, a LoRA-adapted variant, and a training-free EffoVPR-style reranker — evaluated on Pitts30k and MSLS (Part III, Chapters 6–7).

(6) A discussion of results, limitations, and directions for future work (Part III, Chapter 8).

The accompanying GitHub repository [1] contains all implementation code, configuration files, training logs, trained checkpoints, and notebooks, organized so that a reader can reproduce every experiment reported here.

[1] https://github.com/[your-username]/dino-vpr-project

2 The DINO Family: Architecture, Objectives, and Training

This chapter presents a self-contained, graduate-level derivation of the DINO family of models. We begin with the Vision Transformer backbone (Section 2.1), then derive the DINO self-distillation objective and its stabilization mechanisms (Section 2.2), cover the additions introduced in DINOv2 (Section 2.3), and conclude with the contributions of DINOv3 (Section 2.4).

2.1 The Vision Transformer Backbone

The DINO family uses the Vision Transformer [Dosovitskiy et al., 2021] as its backbone. We briefly review it to fix notation.

Given an input image $\mathbf{x} \in \mathbb{R}^{H \times W \times 3}$, ViT splits it into a grid of non-overlapping patches of size $P \times P$ (typically $P = 14$ or 16), yielding $N = HW/P^2$ patches $\{\mathbf{x}_p^{(i)}\}_{i=1}^N$. Each patch is flattened and linearly projected to a d-dimensional token embedding via a learned projection $\mathbf{E} \in \mathbb{R}^{d \times 3P^2}$. A learnable `[CLS]` token $\mathbf{x}_{\text{cls}} \in \mathbb{R}^d$ is prepended, and a learned (DINO, DINOv2) or rotary (DINOv3) positional encoding is added. The resulting sequence

$$\mathbf{z}_0 \;=\; [\mathbf{x}_{\text{cls}};\ \mathbf{E}\mathbf{x}_p^{(1)};\ \ldots;\ \mathbf{E}\mathbf{x}_p^{(N)}] + \mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+1) \times d} \tag{1}$$

is passed through L stacked Transformer blocks, each combining multi-head self-attention (MHSA) and a position-wise feed-forward network (FFN) with pre-LayerNorm and residual connections:

$$\mathbf{z}'_\ell = \mathbf{z}_{\ell-1} + \text{MHSA}\big(\text{LN}(\mathbf{z}_{\ell-1})\big), \tag{2}$$

$$\mathbf{z}_\ell = \mathbf{z}'_\ell + \text{FFN}\big(\text{LN}(\mathbf{z}'_\ell)\big), \quad \ell = 1, \ldots, L. \tag{3}$$

Multi-head self-attention computes, for each of H heads, queries, keys, values $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{(N+1) \times d_h}$ (where $d_h = d/H$) via linear projections of the input, and outputs

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_h}}\right)\mathbf{V}. \tag{4}$$

The final output $\mathbf{z}_L$ provides two kinds of representations used downstream:

- the `[CLS]` token $\mathbf{z}_L^{[\text{cls}]} \in \mathbb{R}^d$, a global image descriptor, and
- the patch tokens $\{\mathbf{z}_L^{(i)}\}_{i=1}^N$, a set of local dense features, one per patch.

For brevity we write $f_\theta(\mathbf{x}) = \mathbf{z}_L^{[\text{cls}]}$ when focusing on the global descriptor.

2.2 DINO: Self-Distillation with No Labels

2.2.1 The Self-Distillation Framework

DINO [Caron et al., 2021] learns representations by matching the output distribution of a *student* network to that of a *teacher* network, where both networks share the same architecture but have different parameters. Crucially, no labels are used; the teacher's output plays the role of a soft target.

Let g_{θ_s} and g_{θ_t} denote the student and teacher respectively, each consisting of a backbone f (a ViT) followed by a projection head h:

$$g_\theta(\mathbf{x}) = h_\theta\big(f_\theta(\mathbf{x})\big) \in \mathbb{R}^K, \tag{5}$$

where K is the output dimension of the projection head (typically $K = 65{,}536$ in the original DINO paper). The projection head is a 3-layer MLP with hidden dimension 2048, GELU

activations, followed by ℓ_2 normalisation and a weight-normalised linear layer that produces the K-dimensional output.

Both networks transform their outputs into probability distributions over the K-dimensional prototype space via temperature-scaled softmax:

$$P_s(\mathbf{x})^{(k)} = \frac{\exp\big(g_{\theta_s}(\mathbf{x})^{(k)}/\tau_s\big)}{\sum_{j=1}^{K}\exp\big(g_{\theta_s}(\mathbf{x})^{(j)}/\tau_s\big)}, \tag{6}$$

$$P_t(\mathbf{x})^{(k)} = \frac{\exp\big((g_{\theta_t}(\mathbf{x})^{(k)} - c^{(k)})/\tau_t\big)}{\sum_{j=1}^{K}\exp\big((g_{\theta_t}(\mathbf{x})^{(j)} - c^{(j)})/\tau_t\big)}, \tag{7}$$

where $\tau_s, \tau_t > 0$ are student and teacher temperatures respectively, and $\mathbf{c} \in \mathbb{R}^K$ is the *center* vector used to avoid mode collapse (derived in Section 2.2.3). Notably $\tau_t \ll \tau_s$, which *sharpens* the teacher's distribution (making it more peaked) while the student's distribution remains relatively smooth — we will see shortly why this asymmetry is essential.

Given two augmented views $\mathbf{x}_1, \mathbf{x}_2$ of the same image, the DINO loss is the cross-entropy between the teacher's distribution on one view and the student's distribution on the other:

$$\mathcal{L}_{\text{DINO}}(\mathbf{x}_1, \mathbf{x}_2) = -\sum_{k=1}^{K} P_t(\mathbf{x}_1)^{(k)} \log P_s(\mathbf{x}_2)^{(k)}. \tag{8}$$

2.2.2 Teacher as EMA of the Student

A crucial design choice in DINO is that the teacher is not a separately trained network: it is constructed as an *exponential moving average* (EMA) of the student's weights. At each iteration t,

$$\theta_t^{(t)} \leftarrow \lambda\, \theta_t^{(t-1)} + (1-\lambda)\, \theta_s^{(t)}, \tag{9}$$

where $\lambda \in [0, 1)$ is a momentum coefficient, typically scheduled from 0.996 to 1.0 over the course of training. No gradient flows into the teacher; its parameters are updated only through Eq. (9). This "momentum teacher" strategy originated in MoCo [He et al., 2020] and BYOL [Grill et al., 2020], and serves two purposes:

1. It provides a *stable*, slowly-evolving target for the student. If the target changed as rapidly as the student's own weights, training would be chasing a moving target and likely diverge.

2. It implements a form of *self-ensembling*: the EMA is effectively an ensemble of past student snapshots, whose averaged representations are empirically higher-quality than any single snapshot.

Caron et al. [2021] show that the teacher consistently outperforms the student throughout training — a property they call the "teacher is better" effect — even though the teacher has no independent learning signal.

2.2.3 Mode Collapse and its Prevention

The loss in Eq. (8) admits trivial solutions:

1. **Uniform collapse:** P_t and P_s are both the uniform distribution over K. The loss reaches $\log K$ and training plateaus with useless features.

2. **Dimensional collapse:** P_t (and consequently P_s) places all mass on a single dimension k^*, independent of the input. The loss is zero (perfectly matched) but the representation carries no information about the input.

Unlike contrastive methods, which prevent collapse through negative pairs, and unlike BYOL, which uses an asymmetric predictor head, DINO prevents collapse using only two operations on the teacher's output: *centering* and *sharpening*.

Centering. The center $\mathbf{c} \in \mathbb{R}^K$ subtracted from the teacher's pre-softmax logits is itself updated as an EMA over the batch:

$$\mathbf{c} \leftarrow m\,\mathbf{c} + (1-m)\,\frac{1}{B}\sum_{i=1}^{B} g_{\theta_t}(\mathbf{x}_i), \tag{10}$$

where $m \in [0,1)$ (typically $m = 0.9$) and B is the batch size. Intuitively, $\mathbf{c}$ tracks the running mean of the teacher's outputs; subtracting it prevents any single dimension from persistently dominating the softmax — it pushes the distribution toward uniformity.

Sharpening. Setting $\tau_t \ll \tau_s$ sharpens the teacher's softmax, pulling its distribution toward a one-hot spike on the most active dimension — the opposite of what centering does.

Why the combination works. Caron et al. [2021] provide a concise justification by decomposing the cross-entropy loss (8) as

$$\mathcal{L}_{\text{DINO}} = H(P_t) + \text{KL}(P_t \,\|\, P_s), \tag{11}$$

where $H(\cdot)$ is the Shannon entropy and KL the Kullback–Leibler divergence. The learning signal for the student comes entirely from the KL term; the entropy term is determined by P_t alone. If $H(P_t) \to 0$ (sharp collapse), the teacher becomes deterministic and further learning is still possible but the representation has collapsed. If $H(P_t) \to \log K$ (uniform collapse), P_t carries no information. Centering and sharpening exert *opposite* pressures on $H(P_t)$: centering raises it, sharpening lowers it. Their balance keeps $H(P_t)$ at an intermediate, informative value where the KL term — and hence the learning signal — remains nontrivial throughout training.

2.2.4 Multi-Crop Augmentation

Rather than feeding a single pair of views to both networks, DINO uses a *multi-crop* strategy [Caron et al., 2020]. From each image, two *global* crops (v_1^g, v_2^g) of size 224×224 (covering $> 50\%$ of the image area) and M *local* crops $(v_1^\ell, \ldots, v_M^\ell)$ of size 96×96 (covering $< 50\%$) are extracted. The teacher processes only the global crops; the student processes both. The full DINO loss is then

$$\mathcal{L}_{\text{DINO}}^{\text{mc}} = \sum_{v^g \in \{v_1^g, v_2^g\}} \sum_{\substack{v' \in V \\ v' \neq v^g}} \text{CE}\big(P_t(v^g) \,\|\, P_s(v')\big), \tag{12}$$

where $V = \{v_1^g, v_2^g, v_1^\ell, \ldots, v_M^\ell\}$ is the full crop set and CE denotes cross-entropy. Conceptually, this forces the student to predict a global view's embedding from a local view of the same image — a "local-to-global" correspondence objective that yields representations encoding strong part–whole relations.

2.2.5 DINO Training Algorithm in Full

Algorithm 1 DINO training (one iteration)

Require: Image batch $\{\mathbf{x}_i\}_{i=1}^{B}$; student θ_s; teacher θ_t; center $\mathbf{c}$; temperatures τ_s, τ_t; momenta λ, m; local crop count M.

1: **for** $i = 1, \ldots, B$ **do**
2: Sample two global crops $v_{1,i}^{g}, v_{2,i}^{g}$ and M local crops $v_{1,i}^{\ell}, \ldots, v_{M,i}^{\ell}$ of $\mathbf{x}_i$.
3: **end for**
4: **Teacher forward (global crops only):** compute $g_{\theta_t}(v_{k,i}^{g})$ for $k \in \{1, 2\}$ (no grad).
5: **Student forward (all crops):** compute $g_{\theta_s}(v)$ for every crop v.
6: **Compute loss** $\mathcal{L}$ via Eq. (12).
7: **Backward and update student:** $\theta_s \leftarrow \theta_s - \eta \nabla_{\theta_s} \mathcal{L}$.
8: **Update teacher (EMA):** $\theta_t \leftarrow \lambda\, \theta_t + (1 - \lambda)\, \theta_s$.
9: **Update center (EMA):** $\mathbf{c} \leftarrow m\, \mathbf{c} + (1 - m)\, \frac{1}{2B} \sum_{i,k} g_{\theta_t}(v_{k,i}^{g})$.

2.2.6 Emergent Properties

A striking empirical finding of Caron et al. [2021] is that the self-attention maps of the final Transformer block in a DINO-trained ViT *implicitly segment* the foreground object — without any segmentation supervision. The `[CLS]` token's attention weights over patch tokens, when visualized as a heatmap, produce masks that are qualitatively comparable to supervised weakly-supervised segmentation methods. This property is absent in supervised ViTs and absent (or much weaker) in contrastive methods like SimCLR. It is one of the strongest pieces of evidence that the DINO objective induces a representation organized around *object semantics* rather than mere discrimination.

2.3 DINOv2: Patch-Level Objectives, Regularization, and Scale

DINOv2 [Oquab et al., 2024] retains the DINO skeleton but introduces several technical modifications — individually modest but collectively transformative — along with a much larger, carefully curated pretraining dataset (LVD-142M).

2.3.1 Patch-Level Objective from iBOT

DINO's loss is computed entirely from the `[CLS]` token and therefore drives the backbone to produce a good *global* representation. Dense downstream tasks (segmentation, depth, correspondence) require high-quality *patch-level* features. DINOv2 addresses this by incorporating the masked-image-modeling objective from iBOT [Zhou et al., 2022].

Let $\mathbf{u}$ be an augmented view of an image, split into patch tokens $\{\mathbf{u}^{(i)}\}_{i=1}^{N}$. Sample a random subset of indices $\mathcal{M} \subset \{1, \ldots, N\}$ to mask. The student receives the *masked* view, in which masked patches are replaced by a learnable mask token [MASK]; the teacher receives the *unmasked* view. Both feed into separate iBOT heads $h_{\theta_s}^{\text{iBOT}}$, $h_{\theta_t}^{\text{iBOT}}$ that produce per-patch K'-dimensional distributions $\tilde{P}_s^{(i)}$, $\tilde{P}_t^{(i)}$. The iBOT loss is

$$\mathcal{L}_{\text{iBOT}} \;=\; -\sum_{i \in \mathcal{M}} \sum_{k=1}^{K'} \tilde{P}_t^{(i),k} \, \log \tilde{P}_s^{(i),k}. \tag{13}$$

The teacher sees the full context and produces clean patch predictions; the student must reconstruct those predictions for the masked patches from the surrounding (visible) tokens. This is a patch-level analogue of the `[CLS]`-level self-distillation in Eq. (8), and it turns out to be critical for dense prediction quality: ablations in Oquab et al. [2024] show roughly 3 mIoU improvement on ADE20K when the iBOT term is included.

2.3.2 Untied Head Weights

DINO, following iBOT, originally used the same projection head for both the `[CLS]` and patch objectives. DINOv2 *unties* them: separate heads h^{DINO} and h^{iBOT} are used for the two losses. This small change delivers measurable gains and reflects the fact that the two objectives require different output spaces.

2.3.3 Sinkhorn–Knopp Centering

DINOv2 replaces the EMA centering of Eq. (10) (for the iBOT teacher head) with Sinkhorn–Knopp (SK) batch normalization [Caron et al., 2020]. Given a batch of teacher outputs $\mathbf{Y} \in \mathbb{R}^{B\times K}$ (after softmax), SK iteratively normalizes $\mathbf{Y}$ so that its row and column sums match prescribed marginals (uniform), yielding a doubly-stochastic matrix. Three SK iterations are typically sufficient. Compared to plain EMA centering, SK normalization is more aggressive at distributing mass across dimensions and better avoids dimensional collapse in the patch-level head.

2.3.4 KoLeo Regularizer

To prevent the feature distribution from clustering too tightly and to encourage a uniform spread of features on the hypersphere, Oquab et al. [2024] add the KoLeo regularizer from Sablayrolles et al. [2019]. For a batch of ℓ_2-normalized `[CLS]` features $\{\hat{\mathbf{z}}_i\}_{i=1}^{B}$, define $d_i = \min_{j\neq i} \|\hat{\mathbf{z}}_i - \hat{\mathbf{z}}_j\|_2$ as the distance from $\hat{\mathbf{z}}_i$ to its nearest neighbor in the batch. The KoLeo loss is

$$\mathcal{L}_{\text{KoLeo}} \;=\; -\frac{1}{B}\sum_{i=1}^{B}\log(d_i). \tag{14}$$

The name derives from the Kozachenko–Leonenko nonparametric estimator of differential entropy: minimising Eq. (14) is equivalent to maximising a lower bound on the entropy of the feature distribution, thereby spreading features out. Empirically, Oquab et al. [2024] report that KoLeo improves instance retrieval accuracy by more than 8% on Oxford-M without harming other metrics — a dramatic improvement for a single loss term.

2.3.5 The Combined DINOv2 Objective

The full DINOv2 training objective is a weighted sum:

$$\mathcal{L}_{\text{DINOv2}} \;=\; \mathcal{L}_{\text{DINO}}^{\text{mc}} \;+\; \alpha_{\text{iBOT}}\,\mathcal{L}_{\text{iBOT}} \;+\; \alpha_{\text{KoLeo}}\,\mathcal{L}_{\text{KoLeo}}, \tag{15}$$

with the coefficients α tuned empirically. All three terms act on the same student, which is trained jointly; the teachers for the DINO and iBOT heads are separate EMAs.

2.3.6 Curated Data and Training-at-Scale Engineering

Beyond objective changes, DINOv2's headline contribution is its curated pretraining dataset LVD-142M (142 million images, built by retrieving images from a large web pool that are close, in embedding space, to images from curated sources like ImageNet-22k and Google Landmarks). The authors argue — and their results show — that curated data is dramatically more sample-efficient than uncurated web crawl: a DINOv2 model trained on 142M curated images outperforms a comparable model trained on a larger uncurated pool.

Training at this scale also demanded engineering work: a custom memory-efficient FlashAttention, Fully Sharded Data Parallel, sequence packing, and stochastic depth. High-resolution adaptation is applied for a short period ($\approx 518 \times 518$) at the end of pretraining, significantly improving performance on dense tasks.

2.4 DINOv3: Gram Anchoring, High-Resolution Refinement, and Beyond

DINOv3 [Siméoni et al., 2025] scales the recipe by another order of magnitude — 1.7B images, a 7B-parameter ViT — but more importantly addresses a newly-identified failure mode of very long SSL schedules.

2.4.1 The Dense Feature Degradation Problem

When the student ViT is trained for very many iterations on massive data, Siméoni et al. [2025] observe an asymmetry: global-feature benchmarks (classification, retrieval) continue to improve, but dense-feature benchmarks (segmentation mIoU, depth RMSE, cosine-similarity-map sharpness) *degrade*. Visually, the patch-to-patch cosine-similarity maps lose locality: a query patch no longer correlates strongly with its spatial neighbors. The underlying cause, as the authors argue, is that the `[CLS]`-oriented DINO loss eventually dominates the patch-level iBOT loss, and patch features begin to "collapse toward" the global representation.

2.4.2 Gram Anchoring

Gram anchoring is a surprisingly elegant remedy. Rather than regularizing the patch features themselves (which might interfere with the main objectives), DINOv3 regularizes their *pairwise similarity structure*.

Let $\mathbf{X}_S \in \mathbb{R}^{N\times d}$ be the student's ℓ_2-normalized patch features for a global crop, with N patches and feature dimension d. The *Gram matrix* $\mathbf{G}_S = \mathbf{X}_S\mathbf{X}_S^\top \in \mathbb{R}^{N\times N}$ records the pairwise patch–patch cosine similarities. Let $\mathbf{X}_G$ be the corresponding patch features produced by a *Gram teacher* — a snapshot of the EMA teacher taken at an earlier checkpoint, *before* the dense-feature degradation set in (typically around iteration 200k). The Gram anchoring loss is

$$\mathcal{L}_{\text{Gram}} \;=\; \left\|\mathbf{X}_S\mathbf{X}_S^\top \;-\; \mathbf{X}_G\mathbf{X}_G^\top\right\|_F^2, \tag{16}$$

i.e. the squared Frobenius norm between the current and reference Gram matrices. The Gram teacher is refreshed periodically (every 10k iterations) to keep pace with the main teacher.

The geometric interpretation is direct: the loss forces the student to preserve the *arrangement* of patches in feature space, even if the features themselves continue to evolve. If early in training patch i was similar to patch j and dissimilar from patch k, that similarity structure is retained. Locality is preserved; global discrimination is not impeded.

2.4.3 High-Resolution Refinement

DINOv3 further uses a *high-resolution Gram teacher*: the Gram teacher processes images at $2\times$ the student's resolution, then downsamples its patch feature map spatially to match the student's patch grid before computing the Gram matrix. This distills the fine spatial structure available at high resolution into the standard-resolution student, giving an additional boost (approximately +2 mIoU on ADE20K).

A subsequent short *high-resolution adaptation* phase mixes global crops at $\{512, 768\}$ and local crops at $\{112, 168, 224, 336\}$ for $\sim$ 10k iterations, improving robustness across inference resolutions — essential for downstream use on high-resolution imagery (remote sensing, medical).

2.4.4 Axial Rotary Positional Embeddings

DINO and DINOv2 used learnable absolute positional embeddings, interpolated at inference for different resolutions. DINOv3 replaces them with axial *rotary positional embeddings* (RoPE) [Su et al., 2024], applied separately to the x and y axes. RoPE is a *relative* encoding: the attention score between two tokens depends only on their relative $(\Delta x, \Delta y)$ displacement, not on absolute positions. This yields far better generalization across input resolutions and aspect ratios.

Siméoni et al. [2025] further introduce *RoPE jittering* (random perturbation of the coordinates during training) as a regularizer.

2.4.5 The Complete DINOv3 Objective and Training Stages

After the main training stage (with loss Eq. (15)) has produced the degraded late checkpoint, a *refinement* stage minimises

$$\mathcal{L}_{\text{DINOv3}}^{\text{refine}} = \mathcal{L}_{\text{DINOv2}} + \alpha_{\text{Gram}} \mathcal{L}_{\text{Gram}} \tag{17}$$

for approximately 100–200k additional iterations. This is followed by high-resolution adaptation and multi-student distillation into smaller variants (ViT-S/B/L/H and a ConvNeXt distillate).

2.4.6 Summary of the Evolutionary Arc

Table 1 summarizes the key differences across the three generations.

Table 1: Evolution of the DINO family. Parameter counts refer to the largest publicly released variant.

	DINO (2021)	**DINOv2** (2023)	**DINOv3** (2025)
Backbone	ViT-B/8 (86M)	ViT-g/14 (∼1.1B)	ViT-7B/16 (6.7B)
Pretrain data	ImageNet-1k	LVD-142M (curated)	LVD-1689M (∼1.7B)
Image-level loss	DINO self-dist. (Eq. 8)	DINO + untied heads	same
Patch-level loss	—	iBOT (Eq. 13)	iBOT
Regularization	Centering + sharpening	+ KoLeo (Eq. 14), Sinkhorn–Knopp	+ Gram anchoring (Eq. 16)
Positional enc.	Learnable abs.	Learnable abs. + hi-res adapt.	Axial RoPE + jittering
Notable property	Emergent attn. seg.	SOTA linear probes	Stable dense features

2.5 Takeaways for This Project

Two observations frame the rest of the report:

1. *The DINO family's design decisions are not ornamental.* Each component — multi-crop, EMA teacher, centering/sharpening, iBOT, KoLeo, Gram anchoring — addresses a specific failure mode. Understanding them is what enables sound choices during adaptation.

2. *Full retraining is not realistic for our project.* DINOv3 was trained with a 7B-parameter ViT on $\sim$ 1.7B images for ∼1M iterations. Our from-scratch exercise (Part II) is therefore deliberately scoped as a pedagogical reimplementation on a small dataset; production-level adaptation uses the publicly released DINOv2 checkpoints as a frozen backbone.

3 Feature Analysis of the Pretrained DINOv2 Backbone

Note. This chapter describes the feature-analysis protocol and the qualitative and quantitative probes we apply to the publicly released DINOv2 backbone. Figures and numerical results are placeholders at this stage and will be filled in with the outputs of notebook `01_dino_feature_exploration.ipynb` once Part II is run. Placeholders are highlighted in gray.

3.1 Objectives

The goals of this chapter are:

(G1) Empirically verify that the pretrained DINOv2 features exhibit the qualitative properties claimed in the literature — semantically coherent attention, clean PCA components, and strong nearest-neighbor structure.

(G2) Diagnose which layers, tokens, and pooling strategies yield the strongest features for (a) global image retrieval and (b) dense patch-level correspondence — directly relevant design choices for our VPR adaptation in Part III.

(G3) Establish quantitative baselines (kNN classification accuracy, zero-shot segmentation mIoU, patch correspondence accuracy) against which we can later measure the effect of adaptation methods.

3.2 Experimental Setup

Model. We use the officially released `facebook/dinov2-base` backbone (ViT-B/14, 86.6M parameters, patch size 14, embedding dimension $d = 768$, $L = 12$ transformer blocks) as our primary model, with `facebook/dinov2-small` (ViT-S/14) used as a smaller-scale control. All experiments use frozen weights — no finetuning, no adapters — since the goal is to characterize the pretrained representation itself.

Datasets. We use three small, publicly available datasets to keep all experiments runnable on a single commodity GPU:

- **ImageNet-100** (100 classes, 50k train / 5k val) for classification probing;
- **PASCAL VOC 2012** validation split for zero-shot segmentation;
- A hand-curated set of ∼50 images sampled from Pitts30k and MSLS for qualitative VPR-relevant visualizations (attention maps, dense feature PCA).

Preprocessing. Images are resized such that the shorter side is 224 pixels (or 518 for high-resolution variants), center-cropped, and normalized with the ImageNet mean and standard deviation.

3.3 Qualitative Probes

3.3.1 Patch-Feature Principal Components

A well-known visualization for DINOv2 features consists in running PCA on the set of patch features $\{\mathbf{z}_L^{(i)}\}_{i=1}^N$ of a single image, retaining the top three components, and interpreting them as an RGB image [Oquab et al., 2024]. If the features are semantically structured, regions belonging to the same object class should map to similar colors.

[Placeholder: 3×3 grid: three query images on the left, their top-3 PCA visualizations of patch features in the middle column, and foreground masks obtained by thresholding the first PCA component on the right.]

Figure 1: Patch-feature PCA visualization (placeholder). **Expected observation:** foreground objects should appear as coherent color blobs distinct from background, and the first PCA component typically separates foreground from background cleanly — a property exploited by several downstream segmentation methods.

3.3.2 Attention Maps of the Final Block

We extract the attention weights from the `[CLS]` token over all patch tokens in each of the H heads of the final Transformer block, visualize them as heatmaps, and compare across heads.

[Placeholder: Grid of attention-map visualizations: for 4 query images, show 6 attention maps (one per head of the final block) overlaid on the input.]

Figure 2: Final-block `[CLS]` attention maps (placeholder). **Expected observation:** in DINO/DINOv2, different attention heads attend to different semantic parts of the image (e.g., one head to the animal's body, another to its head, a third to the ground). This is the emergent-segmentation property discussed in Section 2.2.

3.3.3 Patch-to-Patch Cosine Similarity Maps

For a chosen query patch q, we compute the cosine similarity of $\mathbf{z}_L^{(q)}$ with every other patch $\mathbf{z}_L^{(i)}$ in a *different* image, and visualize the resulting similarity map.

[Placeholder: Pair of images with a query pixel marked on the left image; heatmap overlay on the right image shows cosine similarity of every patch to the query patch. Repeat for 4 query–target pairs.]

Figure 3: Dense patch-to-patch similarity across images (placeholder). **Expected observation:** the similarity map concentrates on the same semantic part in the target image (e.g., a query on a dog's ear lights up the dog's ear in the target). This is directly the mechanism that training-free VPR methods like EffoVPR [Tzachor et al., 2024] exploit for re-ranking.

3.4 Quantitative Probes

3.4.1 Linear and kNN Classification on ImageNet-100

We evaluate global `[CLS]` features by two standard protocols:

- **Linear probe:** train a single linear classifier $\mathbf{W} \in \mathbb{R}^{C \times d}$ on top of frozen `[CLS]` features for 100 epochs with SGD.
- **kNN:** for each validation feature, retrieve the $k = 20$ nearest training features (cosine distance) and vote for the class. No training.

Table 2: Classification on ImageNet-100 with frozen DINOv2 features (placeholder).

Model	Probe	Top-1 (%)	Top-5 (%)
DINOv2-S/14	Linear	*[Placeholder: –]*	*[Placeholder: –]*
DINOv2-S/14	kNN ($k = 20$)	*[Placeholder: –]*	*[Placeholder: –]*
DINOv2-B/14	Linear	*[Placeholder: –]*	*[Placeholder: –]*
DINOv2-B/14	kNN ($k = 20$)	*[Placeholder: –]*	*[Placeholder: –]*

Expected outcome: linear probe will exceed kNN by ∼2–4% top-1. ViT-B will outperform ViT-S by a similar margin. Top-1 accuracy should fall in the 85–90% range, confirming that the frozen features are already highly linearly separable.

3.4.2 Layer-Wise Feature Quality

A practically important question for adaptation is *which layer's features are best* for a given task. We extract `[CLS]` features from every Transformer block $\ell = 1, \ldots, L$ and evaluate each with a kNN probe on ImageNet-100. We repeat the analysis for patch features using a dense cosine-similarity retrieval task.

[Placeholder: Line chart: x-axis = Transformer block index (1–12); y-axis = kNN top-1 (%); two curves, one for global `[CLS]` features and one for pooled patch features.]

Figure 4: Layer-wise feature quality (placeholder). **Expected observation:** `[CLS]`-token quality peaks at the last layer or the penultimate layer; patch features may peak slightly earlier. The gap tells us which layers to tap for the VPR adapter — this directly informs our implementation in Part III.

3.4.3 Zero-Shot Segmentation on PASCAL VOC

We apply k-means clustering ($k = C + 1$, where C is the number of VOC classes) to the set of patch features across the validation split, then match clusters to ground-truth classes via Hungarian assignment and compute mIoU. This is a classical probe for dense-feature quality.

Table 3: Zero-shot segmentation on PASCAL VOC val (placeholder).

Model	Feature extracted from	mIoU (%)
DINOv2-B/14	Last-block patch tokens	*[Placeholder: –]*
DINOv2-B/14	Mean of last 4 blocks	*[Placeholder: –]*
DINOv2-B/14	Final-block K projections	*[Placeholder: –]*

Expected outcome: mid-40s mIoU, with multi-layer averaging yielding 2–3 point improvements over single-layer features. Using internal-attention K-projections (à la EffoVPR) may yield sharper boundaries than direct patch tokens.

3.4.4 Image Retrieval Probe on a Small VPR Sample

As an early diagnostic for our VPR experiments, we take a small subset of Pitts30k queries and references, extract DINOv2 global descriptors (last-block `[CLS]` token, ℓ_2-normalized), and measure Recall@1 on the subset.

Table 4: Zero-shot VPR diagnostic on Pitts30k subset (placeholder).

Descriptor	Recall@1 (%)
DINOv2-B/14 `[CLS]` (last layer)	*[Placeholder: –]*
DINOv2-B/14 GeM-pooled patch tokens (last layer)	*[Placeholder: –]*
DINOv2-B/14 mean-pooled patch tokens	*[Placeholder: –]*

Expected outcome: out-of-the-box features already yield a non-trivial Recall@1 on Pitts30k (literature suggests 60–75% depending on pooling choice) — a strong motivation for our frozen baseline in Part III.

3.5 Planned Analyses and Report-Back

Once Part II is complete, we will return to this chapter and replace all placeholders with the actual figures and numerical values. A brief analytical discussion of each result will be added, including:

- which quantitative observations matched our a priori expectations based on the literature;
- which observations did not, and a hypothesis for why;
- the implications for the VPR adaptation choices in Part III.

3.6 Summary of Part I

We have introduced the setting and motivation (Chapter 1), derived the DINO family in full mathematical detail (Chapter 2), and laid out the feature-analysis protocol we will execute in Part II (this chapter). Part II will add: the mini-reimplementation of the DINO training loop on CIFAR-100 / Tiny-ImageNet, and the numerical completion of the present chapter. Part III will add the VPR adaptation experiments and the empirical comparison of multiple adaptation strategies.

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